

🕒 Social Sandbox: 24-Hour Attack Plan (Local API Edition)

Teammate 1 (The Brain): Local Flask Backend, Vultr/Ollama Dumb Engine Setup, AI Prompt Engineering, API Routing. **Teammate 2 (The Body):** Hybrid Frontend (HTML/Tailwind + Vue Chat), MongoDB Atlas, Client-side ML (`face-api.js` , Hark.js, Web Speech).

Golden Rule: If a feature takes more than 2 hours and isn't working, mock the data and move on. The judges care about the *demo*, not backend perfection.

Time (EST)	Phase	Teammate 1 (Local Backend / AI)	Teammate 2 (Frontend / Client ML)
12:00 PM	🚀 Kickoff & Setup	Initialize Git repo. Deploy Vultr GPU Instance.	Scaffold HTML/Tailwind pages. Create MongoDB Atlas cluster.
1:00 PM	☁️ The 5-Min Cloud	Vultr Speedrun: SSH in, install Ollama, expose port 11434. <i>Done with Vultr.</i>	Build <code>index.html</code> (Login) and <code>lounge.html</code> (Dashboard).
2:00 PM	🔥 Local Hello World	Setup local Flask <code>app.py</code> . Create <code>/ping</code> and verify connection to Vultr IP.	Connect PyMongo to Flask. Create dummy user document in Atlas.
3:00 PM	🗃️ Comm Suite (Chat)	Write the 4 distinct LLaMA 3 system prompts (Sarcasm, Icebreaker, Hint, Ping-Pong).	Build <code>chat-module.html</code> . Mount Vue via CDN for the chat UI state.
4:00 PM	🗃️ Comm Suite (API)	Build <code>/api/analyze_chat</code> route to swap prompts based on frontend requests.	Finish Vue logic for rendering chat bubbles and tracking user input.
5:00 PM	💛 Integration 1	[SYNC] Connect the Vue chat component to Flask. Verify all 4 chat modules work using the single Vue UI.	
6:00 PM	🍕 Dinner Break	Eat. Step away from the screen. Discuss the Audio module strategy.	Eat. Hydrate.
7:00 PM	🎙️ Vocal Sandbox	Setup ElevenLabs API. Create <code>/api/generate_audio</code> and <code>/api/analyze_speech</code> .	Implement native Web Speech API and Hark.js in <code>module-audio.html</code> .

8:00 PM	 Vocal Sandbox	Build the pipeline: STT Transcript -> LLaMA Grading -> ElevenLabs TTS.	Build "Walkie-Talkie" UI. Hook up mic logic and hesitation detection.
9:00 PM	 Integration 2	[SYNC] Connect audio output to Flask. Play ElevenLabs audio file back in the UI and show Confidence/Stutter score.	
10:00 PM	 Facial Gym	Build Flask logic for "Expression Reader" quiz and XP logging.	Download <code>face-api.js</code> weights. Setup <code><video></code> and <code><canvas></code> in <code>module-face.html</code> .
11:00 PM	 Facial Gym	Refine LLaMA prompts across all modules to be faster/more accurate.	Write the <code>face-api.js</code> loop. Track expressions for the "Face Emotions Helper" and trigger UI feedback.
12:00 AM	 The Midnight Grind	Write fallback/mock JSON responses in Flask in case Vultr dies during demo.	Ensure XP from all 3 "Gyms" correctly triggers the Flask database update.
1:00 AM	 Bug Squashing	Fix CORS errors, handle ElevenLabs API rate limits/timeouts.	Fix CSS bugs, ensure Flexbox/Grid layouts don't break on different screens.
2:00 AM	 End-to-End Test	[SYNC] Play through the entire app as a user. Log every bug in a text file.	
3:00 AM	 Triage	Fix <i>only</i> the bugs that will break the demo. Ignore minor aesthetic issues.	Fix <i>only</i> the bugs that will break the demo.
4:00 AM	 Polish	Fine-tune the "Social Coach" persona so the AI sounds professional and empathetic.	Add Accessibility features (ARIA labels, contrast checks, hover states).
5:00 AM	 Buffer / Nap	<i>If on schedule:</i> Sleep. <i>If behind:</i> Hardcode the broken parts to fake it.	<i>If on schedule:</i> Sleep. <i>If behind:</i> Hardcode the broken parts to fake it.
6:00 AM	 Buffer / Nap	(Continue Buffer/Nap)	(Continue Buffer/Nap)
7:00 AM	 Morning Prep	Draft Devpost submission (inspiration, tech stack, what we learned).	Final UI polish. Make sure the MongoDB Atlas data looks clean/realistic.

8:00
AM

 **BACKUP
VIDEO**

[CRITICAL] Record a 2-minute screencast of the app working perfectly. (Wi-Fi always fails during judging).

9:00
AM

 **Pitch
Practice**

Rehearse pitch. **Teammate 1:** Talk about Vultr dumb-engine flex and prompt engineering.

Rehearse pitch. **Teammate 2:** Talk about MongoDB, Vanilla/Vue architecture, and Client-side ML.

10:00
AM

 **CODE
FREEZE**

DO NOT TOUCH THE CODE. Ensure Flask is running locally without errors.

DO NOT TOUCH THE CODE. Ensure browser permissions (mic/cam) are granted.